

# Vulnerability Scanner Installation & Configuration

## OpenVAS / GVM — Security Assessment Report

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COURSE: **Managing Vulnerabilities**

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CLASSIFICATION: **Educational Use Only**

Exec. Executive Summary

This lab documents the installation and configuration of an automated vulnerability scanner in a controlled, isolated virtual environment. OpenVAS/GVM was installed on a Kali Linux scanner VM, configured with a Debian target VM, and used to initiate an initial vulnerability assessment. Findings were to be reviewed by severity using CVSS scores, and one notable vulnerability selected for validation against the NVD database.

**Note:** The scan could not be completed due to a UI issue in GVM where the scan button remained inactive. This is documented transparently in Section 4. All prior configuration steps were completed successfully.

1. Lab Environment

| Component               | Details                    |
|-------------------------|----------------------------|
| Scanner VM              | Kali Linux — GVM/OpenVAS   |
| Scanner Host-only IP    | 192.168.56.111/24 (eth1)   |
| Scanner NAT IP          | 10.0.2.15/24 (eth0)        |
| Target VM               | Debian (Lab target)        |
| Target Host-only IP     | 192.168.56.110/24 (enp0s8) |
| Target NAT IP           | 10.0.2.15/24 (enp0s3)      |
| Virtualization Platform | VirtualBox / VMware        |
| Host-only Network       | 192.168.56.0/24            |
| NAT Network             | 10.0.2.0/24                |

Scanner VM  
192.168.56.111  
Kali Linux / GVM

■ Host-only 192.168.56.0/24 ■

Target VM  
192.168.56.110  
Debian

Network topology — Host-only isolated virtual network

2. Installation Notes

The scanner VM was updated and GVM was installed using the system package manager. The automated setup process initialized the PostgreSQL database, synchronized the vulnerability test (NVT) feeds from Greenbone Community Feed, and generated administrative credentials.

Commands executed in sequence:

| Step | Description | Command / URL |
|------|-------------|---------------|
|------|-------------|---------------|

|   |                      |                                                             |
|---|----------------------|-------------------------------------------------------------|
| 1 | System update        | <code>sudo apt update</code>                                |
| 2 | System upgrade       | <code>sudo apt upgrade -y</code>                            |
| 3 | GVM installation     | <code>sudo apt install gvm -y</code>                        |
| 4 | GVM database init    | <code>sudo gvm-setup</code>                                 |
| 5 | Start GVM services   | <code>sudo gvm-start</code>                                 |
| 6 | Verify configuration | <code>sudo gvm-check-setup</code>                           |
| 7 | Web interface (GSA)  | <a href="https://127.0.0.1:9392">https://127.0.0.1:9392</a> |



**Setup Result:** GVM setup completed successfully. Admin credentials were generated automatically. NVT feeds synchronized from Greenbone Community Feed. OpenVAS default scanner confirmed active on socket.

## 3. Configuration Choices

Configuration decisions directly affect scan accuracy and scan duration. For this lab, a safe baseline policy was chosen to balance coverage and stability.

### 3.1 Target Configuration

Navigation path: **Configuration** → **Targets** → **New Target**

| Parameter   | Value                                                      |
|-------------|------------------------------------------------------------|
| Target Name | Debian-Lab7                                                |
| Host        | 192.168.56.110                                             |
| Port List   | All IANA Assigned TCP (or OpenVAS Default for faster scan) |
| Alive Test  | Scan Config Default                                        |

### 3.2 Scan Configuration (Policy)

| Setting          | Value           | Rationale                                                |
|------------------|-----------------|----------------------------------------------------------|
| Scan Config      | Full and fast   | Broad NVT coverage in reasonable lab timeframe           |
| Scan Type        | Unauthenticated | Default; authenticated adds depth if SSH creds available |
| Concurrent Tasks | Default (1)     | Safe for lab; avoids overloading target VM               |

### 3.3 Optional: Authenticated Scan Credentials

If SSH access is available on the target, credentials can be configured to increase scan depth and accuracy, enabling GVM to perform local checks on installed packages.

Navigation path: **Configuration** → **Credentials**

| Field    | Example Value (Metasploitable2) |
|----------|---------------------------------|
| Type     | SSH                             |
| Username | msfadmin                        |
| Password | msfadmin                        |
| Port     | 22                              |

#### 4. Running the Scan & Reviewing Results

 **Issue Encountered:** The scan could not be initiated because the Start Scan button in the GVM web interface (Greenbone Security Assistant) was inactive / greyed out. The root cause could not be determined during the lab session. As a result, Sections 4 and 5 could not be completed with live scan data.

**Likely causes for inactive scan button:**

- Feed synchronization still in progress — GVM blocks scanning until NVT feeds are fully loaded
- GVM services not fully started — ospd-openvas socket may not have been ready
- Task not properly saved / missing required fields before attempting to run
- Resource constraints on the VM (RAM/CPU) causing the ospd-openvas daemon to be unresponsive

**Recommended troubleshooting steps:**

| Action            | Command                                                 |
|-------------------|---------------------------------------------------------|
| Check feed status | <code>sudo runuser -u _gvm -- greenbone-nvt-sync</code> |
| Verify services   | <code>sudo gvm-check-setup</code>                       |
| Restart all       | <code>sudo gvm-stop &amp;&amp; sudo gvm-start</code>    |
| Check daemon      | <code>sudo ss -lpx   grep ospd</code>                   |

#### 5. Expected Findings (Reference — Scan Not Completed)

Although live scan results were not obtained, the following represents the type of findings typically produced by GVM against a Debian or Metasploitable2 target, organized by CVSS severity for reference:

| Severity | CVSS Range | Example Finding Type                              | Action                       |
|----------|------------|---------------------------------------------------|------------------------------|
| Critical | 9.0–10.0   | Remote Code Execution (RCE) via unpatched service | Immediate patch / isolation  |
| High     | 7.0–8.9    | Outdated SSH / Apache with known CVEs             | Patch or mitigate within 72h |
| Medium   | 4.0–6.9    | Default credentials, weak cipher suites           | Harden configuration         |
| Low      | 0.1–3.9    | Information disclosure, verbose banners           | Review and remediate         |

|      |     |                                            |                    |
|------|-----|--------------------------------------------|--------------------|
| Info | 0.0 | Open ports, OS detection, service versions | Document & monitor |
|------|-----|--------------------------------------------|--------------------|

## 6. Conclusion

This lab successfully covered the installation and configuration phases of deploying an enterprise-grade vulnerability scanner (OpenVAS/GVM) in an isolated virtual environment. The following steps were completed:

| Step | Action                                | Status                    |
|------|---------------------------------------|---------------------------|
| 1    | VM network setup & verification       | Completed                 |
| 2    | GVM installation (apt + gvm-setup)    | Completed                 |
| 3    | Feed synchronization                  | Completed                 |
| 4    | Target configuration (Debian-Lab7)    | Completed                 |
| 5    | Scan policy selection (Full and fast) | Completed                 |
| 6    | Scan execution                        | Blocked — button inactive |
| 7    | Results review & NVD validation       | Not reached               |

|             |                                                                                                                                                                                                                                                    |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| KEY FINDING | The installation and configuration of GVM was fully successful. The inability to execute the scan was a technical/UI issue unrelated to the student's configuration work. Troubleshooting steps have been identified and documented for follow-up. |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|            |                                                                          |
|------------|--------------------------------------------------------------------------|
| Tools Used | GVM / OpenVAS (Greenbone)   GSA Web Interface   nmap (host verification) |
|------------|--------------------------------------------------------------------------|